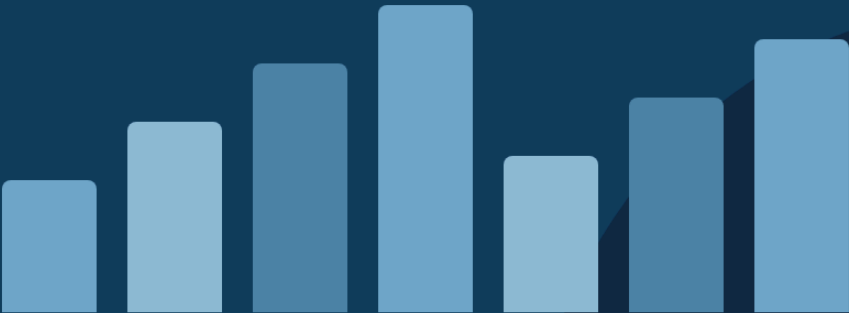


N O R T H L I G H T



Northlight Weather API

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Northlight Weather API v1

Northlight serves current conditions and short-range forecasts for a network of weather stations. This handbook is one set of Markdown documents that cudoc publishes three ways: as the site you are reading, as a bound PDF with a cover and contents page, and as a Word file with real styles. Every figure below is drawn from the reference pages, so a correction there reaches this page too.

How this handbook is built

The site, the PDF and the Word file come from one `cudoc-export build run` and one set of design tokens. The configuration is a single file, `showcase.config.mjs`, beside the documents.

Endpoints at a glance

Each row is one section of the endpoint reference. The method and the path are read from the table at the top of that section, and the description is its first paragraph.

Endpoint	Method	Path	What it returns
Current conditions	GET	/v1/weather/current	Returns the latest observation from one station.
Hourly forecast	GET	/v1/forecast/hourly	Returns up to 48 hourly forecast points for one station.
Station directory	GET	/v1/stations	Lists the stations a key may read, with their location and status.

Where to start

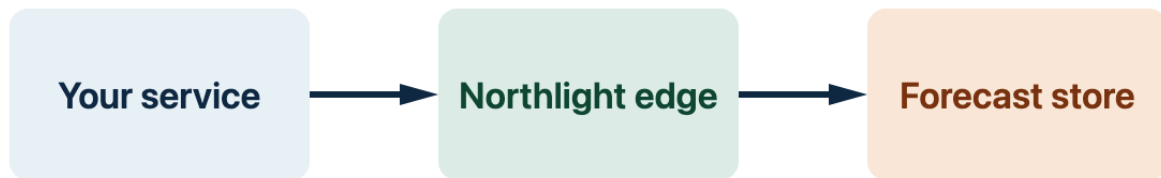
1. Read [Getting started](#) for a key and a first request.
2. Keep [Rate limits](#) open while you design a polling schedule.
3. Check the [release notes](#) before upgrading a client.

Reading this on paper?

External links print their address after the text, and links between documents jump inside the bound file.

Getting started

Three steps take you from nothing to a first forecast. The whole flow is one HTTPS request with a key in a header; there is no SDK to install.



GET /v1/weather/current?station=... → cached at the edge for 60 seconds → read from the store on a miss

Get a key

Keys are issued per project and scoped to the stations a project may read.

Requirements

Step	Prerequisites
Register	• A project name • A billing contact • At least one station • Accepted terms • A callback URL
Rotate	• The current key • Owner role

Keep the key out of the browser

A key grants every scope its project holds. Call Northlight from your own backend and pass results on; never ship the key in client-side code.

Make a request New

Send the key in the `Authorization` header and ask for a station by its code:

```
curl -H "Authorization: Bearer $NORTHLIGHT_KEY" \  
  "https://api.northlight.example/v1/weather/current?station=OSL-01"
```

```
{  
  "station": "OSL-01",  
  "observedAt": "2026-09-22T06:00:00Z",  
  "temperatureC": 11.4,  
  "windMs": 3.2,  
  "conditions": "overcast"  
}
```

Every response carries `observedAt`, the instant the station reported¹. A missing station returns `404` with a machine-readable `code`, listed under [Error codes](#).

Stay within the limits

Current conditions are cached at the edge for sixty seconds, so polling faster than that returns the same reading and spends your quota. The windows and the headers that report them are in [Rate limits](#); the short version is below.

Windows

As a new integrator you share one rule: a key may make 60 requests per rolling minute to `/v1/weather/current` and 600 per rolling hour across all endpoints. Both counters are reported on every response.

Header	Meaning
<code>X-Limit-Minute</code>	Requests left in the current minute window.
<code>X-Limit-Hour</code>	Requests left in the current hour window.
<code>Retry-After</code>	Seconds to wait, present only on a <code>429</code> .

Bursts are counted per key

Two services sharing one key share one window. Give each service its own key so a spike in one cannot starve the other.

Footnotes

1. Always UTC, always with a **Z** suffix. Convert at the edge of your own system, not in the data layer. ↩

Endpoint reference

Each section describes one endpoint. The first table of a section is its basic information, which the overview page reads into a summary table; the tables that follow describe parameters and the response. Error responses are listed separately under [Error codes](#).

Current conditions

Returns the latest observation from one station.

Method		Path		Auth
GET		/v1/weather/current		Bearer
Parameter	Type	Required	Description	
station	string	yes	The station code, such as 0SL-01 . One station per request.	
units	string	no	metric (default) or imperial .	

The response is one observation object. Readings older than fifteen minutes carry stale: true .

Hourly forecast

Returns up to 48 hourly forecast points for one station.

Method		Path	Auth
GET		/v1/forecast/hourly	Bearer
Parameter	Type	Required	Description
station	string	yes	The station code.
hours	integer	no	1 to 48, default 24. Values above 48 are clamped, not rejected.
fields	list	no	- temperature - wind - precipitation - Probability and amount

Station directory

Lists the stations a key may read, with their location and status.

Method		Path		Auth
GET		/v1/stations		Bearer
Parameter	Type	Required	Description	
region	string	no	An ISO 3166-2 code to narrow the list.	
status	string	no	active , maintenance or retired .	

Rate limits

Limits protect the stations, which report on fixed schedules, from being asked the same question thousands of times a minute. They are generous for any client that respects the cache.

Windows

Every plan shares one rule: a key may make 60 requests per rolling minute to `/v1/weather/current` and 600 per rolling hour across all endpoints. Both counters are reported on every response.

Header	Meaning
<code>X-Limit-Minute</code>	Requests left in the current minute window.
<code>X-Limit-Hour</code>	Requests left in the current hour window.
<code>Retry-After</code>	Seconds to wait, present only on a <code>429</code> .

Backoff

On a `429` or a `503`, wait for `Retry-After` when present, otherwise double the delay from one second up to thirty, and stop after five attempts.

Plans compared

The table has seven columns, so the paginated outputs put it on a landscape page of its own.

Plan	Stations	Minute window	Hour window	Forecast hours	History	Support
Community	3	60	600	24	7 days	Forum
Team	25	120	6,000	48	90 days	Email, 2 days
Business	250	600	60,000	48	1 year	Email, 1 day
Enterprise	Custom	Custom	Custom	48	Custom	Named contact

Appendix: measuring your own usage

Log the two limit headers with every response and chart the minimum per minute. A flat line near zero means a client is polling faster than the sixty-second cache refreshes; slow it down rather than raising the plan.

Retired keys keep counting

Requests made with a retired key are rejected but still counted against the project for one hour, so rotate keys during a quiet period.

Error codes

Every error response carries a `code` from this table and a human-readable `message` that may change between releases; branch on the code, never on the message.

Code	HTTP	Meaning	Retry
<code>station_unknown</code>	404	No station has that code.	No
<code>station_retired</code>	410	The station stopped reporting permanently.	No
<code>key_scope</code>	403	The key's project does not cover the station.	No
<code>rate_limited</code>	429	The window is exhausted.	After <code>Retry-After</code>
<code>upstream_delayed</code>	503	The station has not reported in time.	With backoff

Retrying

The backoff rule for `rate_limited` and `upstream_delayed` is in [Rate limits](#).

Release notes

Changes to the API, newest first. Dates are the day a change reached every region.

2026-09-15 Latest

- `fields` on the hourly forecast accepts `precipitation`, returning probability and expected amount. → [Hourly forecast](#)
- `X-Limit-Hour` is now sent on every response, not only when the minute window is low.

2026-07-02

- Readings older than fifteen minutes carry `stale: true` instead of being withheld.
- The `station_retired` error replaces a generic `404` for stations that stopped reporting permanently.

2026-04-10

- First public release of `/v1` .

Deprecation policy

A field or endpoint is announced here at least ninety days before it changes shape, and the previous behaviour stays available for that period.